Noise-Robust Radar HRR Target Recognition Based on Complex Gaussian Statistical Models

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Abstract: In order to utilize the phase information to improve radar target recognition performance of high range resolution (HRR) target signals, this paper takes adaptive Gaussian classifier (AGC) model and factor analysis (FA) model for instance to generalize Gaussian statistical models to complex domain to build complex HRR models. It is demonstrated that the structures and parameter estimators of complex Gaussian statistical models are invariant to the initial phases of HRR complex data. Furthermore, to enhance the recognition performance under low signal-to-noise ratio (SNR) conditions, a noise-robust modification algorithm is introduced. It solves the mismatch between complex Gaussian models and noisy test signals. Experimental results show that the proposed models can obtain higher average correct recognition rates by utilizing the phase information. Also, the modified models can deal well with noisy test signals.

Keywords: high range resolution (HRR); radar target recognition; complex Gaussian distribution; noise robustness

I. INTRODUCTION

The high range resolution (HRR) radar target signal is the coherent summations of the complex time return from target scatterers in each range cell, which contains abundant target information, such as target size, scatterer intensity etc [1-3]. At present, the recognition methods based on HRR target signals mainly use the real profiles. In this way the recognition performance will be limited, since the phase information of HRR target signals, as well as the amplitude information, reflects the target structure. However, the initial phase caused by translation motion in an HRR signal is extremely sensitive to the variance of target distance. Therefore, it is difficult to utilize the phase information via some traditional recognition methods. In this paper, we take adaptive Gaussian classifier (AGC) [4] and factor analysis (FA) [5-7] for instance to generalize Gaussian models to complex domain and adopt the complex Gaussian models to develop complex models for HRR complex data. We demonstrate the structures and parameter estimators of complex Gaussian models are invariant to the initial phase, and the phase information can be utilized to improve recognition performance.

In addition, an adaptive noise-robust modification algorithm is introduced. In the application of radar target recognition, the training signals usually have high signal-to-noise ratio (SNR), while the SNR of test signals is much lower for the target at a long distance. Thus the recognition performance of the test signals under low SNR will greatly worsen because of the SNR mismatch between learned models and test signals [2-5]. In order to get robust recognition performance, the model parameters learned in the training stage should be modified in the test stage. The noise component in a test complex-value HRR target signal can usually be assumed as the additive Gaussian noise, where the test signal can be expressed as a linear model, *i.e.*, $\hat{\mathbf{x}} = \mathbf{x} + \mathbf{n}$ with $\mathbf{n}$ denoting the complex-valued noise, $\mathbf{x}$ denoting the complex-valued signal under high SNR, and $\hat{\mathbf{x}}$ denoting the complex-valued noisy signal [1]. Based on such a linear structure, the model parameters for $\hat{\mathbf{x}}$ can be easily and accurately derived in complex domain. However, the absolute value operator to obtain the real-valued signal completely changes the linear structure, *i.e.*, $|\hat{\mathbf{x}}| = |\mathbf{x} + \mathbf{n}| \neq |\mathbf{x}| + |\mathbf{n}|$. Thus it is difficult to accurately get the model parameters for $\hat{\mathbf{x}}$ in the real domain to match the variation of SNR. Based on some approximations, papers [4,5] propose noise-robust modification algorithms in the real signal domain to solve the SNR mismatch problem. The experimental results show a gain of recognition rate under the test condition of low SNR. Nevertheless, the approximated modification algorithms cannot yield good recognition performance when $\mathrm{SNR} \le 15\mathrm{dB}$. In addition, since the modification algorithms in [4,5] involve some complicated computation in the test stage, their recognition speeds decrease. In this paper, we propose an noise-robust modification algorithm to modify the parameters of complex Gaussian models directly in the complex domain. In this way, the statistical characteristics of test signals match with the modified models under low SNR condition. The experimental results on measured data show clearly that the noise-robust modification algorithm can efficiently improve the recognition performance.

II. COMPLEX GAUSSIAN MODELS AND NOISE-ROBUST MODIFICATION ALGORITHM

A. Complex Gaussian Models

HRR signals are complex vectors which are the summations of target scatterer echoes onto the radar line-of-sight [1-5]. As well as the amplitude information, the phase information which can be utilized to improve the recognition performance, reflects target physical structure information [1].

Suppose there are radial displacement and little uniform rotation for a target. As discussed in reference [5], the returned HRR complex signal is denoted as a discrete complex vector, the nth complex returned echo from the lth range cell $(l = 1, 2, \cdots, L)$ in baseband can be approximated as

$$x_{nl} = e^{j\theta_n} \sum_{i=1}^{V_l} \sigma_l(i) e^{j\phi_{nl}(i)} \tag{1}$$

Where $\sigma_l(i)$ represents the amplitude of the ith scatterer in the lth range cell, V_l denotes the number of target scatterers in the lth range cell, θ_n denotes the initial phase of the nth returned echo, which depends on the target distance and the radar wavelength and does not contain the target discriminative information, and $\phi_{nl}(i)$ denotes the remained echo phase of the ith scatterer in the lth range cell of the nth returned echo.

Then the nth HRR complex signal is

$$\mathbf{x}_n = [x_{n1}, x_{n2}, \cdots, x_{nL}]^T = e^{j\theta_n}[\tilde{x}_{n1}, \tilde{x}_{n2}, \cdots, \tilde{x}_{nL}]^T \qquad (2)$$

Where $\tilde{x}_{nl} = \sum_{i=1}^{V_l} \sigma_{li} e^{j\phi_{nl}(i)}$ with $l = 1, 2, \cdots, L$. Let λ denotes the radar wavelength, and R_n represents the radial distance between the target reference center in the nth returned echo and the radar. Here the initial phase of the nth HRR complex sample, *i.e.*, $\theta_n = -\frac{4\pi}{\lambda} R_n$, is strongly sensitive to the target position variation. For example, for carrier frequency of 5520 MHz (approximately, the wavelength of 5 cm), if the radial displacement of a target is 1.25 cm, the initial phase difference will be π. In application, the measurement precision of the target distance usually cannot satisfy the initial phase compensation , therefore, the measured HRR complex signals have the initial phase sensitivity. Next we analyze the effect of initial phase variation to HRR statistical models under the hypothesis of complex Gaussian distribution.

Based on the complex Gaussian distribution assumption, HRR complex signals obey the complex Gaussian distribution with zero-mean [1]. A $P \times 1$ dimensional HRR complex sample $\mathbf{x}$ is given, $\mathbf{x} \sim CG(\mathbf{x} \mid \mathbf{0}, \mathbf{\Sigma})$, where $\mathbf{\Sigma}$ is covariance matrix. The probability density function can be shown as:

$$p(\mathbf{x}) = \pi^{-P} \mid \mathbf{\Sigma} \mid^{-1} \exp\left(\mathbf{x}^H \mathbf{\Sigma}^{-1} \mathbf{x}\right) \qquad (3)$$

Where H denotes the conjugate transpose operation. Let $\mathbf{x}$ multiplies the random initial phase *i.e.* $\mathrm{e}^{j\theta}$, and we obtain a new HRR sample $\mathbf{x}' = \mathbf{x}\,\mathrm{e}^{j\theta}$. According to the linear properties of complex Gaussian distribution, $\mathbf{x}'$ still obeys the complex Gaussian distribution with zero-mean. let $\mathbf{\Sigma}'$ represents the new covariance, then:

$$\begin{aligned}\mathbf{\Sigma}' &= E\left(\mathbf{x}'\mathbf{x}'^H\right) = E\left[\left(\mathbf{x}\,\mathrm{e}^{j\theta}\right)\left(\mathbf{x}\,\mathrm{e}^{j\theta}\right)^H\right] \\ &= E\left[\left(\mathbf{x}\,\mathrm{e}^{j\theta}\right)\left(\mathbf{x}^H\,\mathrm{e}^{-j\theta}\right)\right] = E\left(\mathbf{x}\mathbf{x}^H\right) = \mathbf{\Sigma}\end{aligned} \qquad (4)$$

Equation (4) indicates $\mathbf{x}'$ obey the same complex Gaussian distribution with zero-mean as $\mathbf{x}$, *i.e.*, $\mathbf{x}' \sim CG(\mathbf{x} \mid \mathbf{0}, \mathbf{\Sigma})$. Under the hypothesis of complex Gaussian distribution, the whole recognition information is contained via the covariance matrix of HRR complex data. The initial phase does not affect the covariance of HRR complex data, *i.e.*, the complex Gaussian models' parameter estimators do not vary with the initial phase.

In the following, we take complex adaptive Gaussian classifier (CAGC) and complex factor analysis (CFA) model for instance to build complex models for HRR data. Similar to AGC [4], CAGC model supposes each range cell echo is mutually independent, and a sample obeys complex Gaussian distribution with zero-mean, *i.e.*, $\mathbf{x} \sim CG(\mathbf{x} \mid \mathbf{0}, \mathbf{\Sigma})$, where $\mathbf{\Sigma} = \mathbf{\Lambda} \in R^{P \times P}$. Since each range cell echo is mutually independent, the matrix $\mathbf{\Lambda}$ is diagonal, and the off-diagonal elements of $\mathbf{\Lambda}$ are zero. Given a frame dataset $\mathbf{X} = [\mathbf{x}_n]_{n=1}^N$, the covariance matrix $\mathbf{\Sigma}$ can be estimated by maximum likelihood:

$$\mathbf{\Sigma} = \mathbf{\Lambda} = \frac{1}{N} diag\left(\sum_{n=1}^{N} \mathbf{x}_n \mathbf{x}_n^H\right) \qquad (5)$$

The operator $diag(\cdot)$ denotes zeroing the off-diagonal elements.

Reference [5] proposed FA model based on multitask learning in the real domain. In order to utilize the phase information, next we adopt CFA model for HRR complex samples. Similar to FA, our CFA assumes HRR signals obey joint Gaussian distribution, and the model can be expressed as:

$$\mathbf{x} = \mathbf{A}\mathbf{z} + \boldsymbol{\varepsilon} \qquad (6)$$

Where $P \times 1$ dimensional $\mathbf{x}$ is a HRR sample; $\mathbf{A} \in R^{P \times K}$ represents loading matrix ($P > K$); $K \times 1$ dimensional complex factor $\mathbf{z} \sim CG(\mathbf{z} \mid \mathbf{0}, \mathbf{I}_P)$, with $\mathbf{I}_P$ being a P dimensional identical matrix; $P \times 1$ dimensional complex vector $\boldsymbol{\varepsilon}$ denotes model noise with $\boldsymbol{\varepsilon} \sim CG(\boldsymbol{\varepsilon} \mid \mathbf{0}, \mathbf{\Psi})$ and the covariance $\mathbf{\Psi}$ being diagonal. From the linear properties of complex Gaussian distribution, we can obtain $\mathbf{x} \sim CG(\mathbf{x} \mid \mathbf{0}, \mathbf{A}\mathbf{A}^H + \mathbf{\Psi})$.

Now we extend expectation maximization algorithm (EM) for FA model in [8,9] to the complex domain to estimate the parameters of CFA model. Given the observed dataset by $\mathbf{X} = [\mathbf{x}_n]_{n=1}^N$, the EM algorithm maximizes the expected complete log likelihood $Q(\mathbf{\Theta})$, which can be written:

$$\begin{aligned}Q(\mathbf{\Theta}) &= E\left\{\sum_{n=1}^{N} \ln\left[p(\mathbf{x}_n, \mathbf{z}_n)\right] \middle| \mathbf{X}\right\} \\ &= E\left\{\sum_{n=1}^{N} \ln\left\{\frac{1}{(\pi)^{(P+K)} \mid \mathbf{\Psi} \mid} \exp\left[-\mathbf{S}\right]\right\} \middle| \mathbf{X}\right\} \\ &= -\sum_{n=1}^{N} \left\{\begin{array}{l}\mathbf{x}_n{}^H \mathbf{\Psi}^{-1} \mathbf{x}_n - 2\mathbf{x}_n{}^H \mathbf{\Psi}^{-1} \mathbf{A} E(\mathbf{z}_n | \mathbf{x}_n) \\ + Tr\left[(\mathbf{A}^H \mathbf{\Psi}^{-1} \mathbf{A} + \mathbf{I}_K) E(\mathbf{z}_n \mathbf{z}_n{}^H | \mathbf{x}_n)\right]\end{array}\right\} \\ &\quad - N(P+K)\ln(\pi) - N \ln \mid \mathbf{\Psi} \mid; \\ \mathbf{S} &= (\mathbf{x}_n - \mathbf{A}\mathbf{z}_n)^H \mathbf{\Psi}^{-1} (\mathbf{x}_n - \mathbf{A}\mathbf{z}_n) + \mathbf{z}_n{}^H \mathbf{z}_n\end{aligned} \qquad (7)$$

In (7), $\mathbf{\Theta} = \{\mathbf{A}, \mathbf{\Psi}\}$ denotes the parameter set of CFA; $p(\mathbf{x}_n, \mathbf{z}_n)$ denotes the joint probability density; and $Tr(\cdot)$ represents to calculate the trace of a matrix. The conditional mean and correlation matrixes of $\mathbf{z}_n$ can be written as:

$$\begin{aligned}&E[\mathbf{z}_n \mid \mathbf{x}_n, \mathbf{\Theta}] = \mathbf{A}^H \mathbf{C}^{-1} \mathbf{x}_n \\ &E[\mathbf{z}_n \mathbf{z}_n{}^H \mid \mathbf{x}_n, \mathbf{\Theta}] = \mathbf{I}_K - \mathbf{A}^H \mathbf{C}^{-1} \mathbf{A} + E[\mathbf{z}_n \mid \mathbf{x}_n, \mathbf{\Theta}] E[\mathbf{z}_n{}^H \mid \mathbf{x}_n, \mathbf{\Theta}]\end{aligned} \qquad (8)$$

Where $\mathbf{C}=\mathbf{A}\mathbf{A}^H+\boldsymbol{\Psi}$. Correspondingly, in the $(\tau+1)$th time of the EM algorithm, the model parameter updating equations are inferred as:

$$\begin{aligned}\tilde{\mathbf{A}}(\tau+1)&=\left[\sum_{n=1}^{N}\mathbf{x}_n E(\mathbf{z}_n{}^H\mid\mathbf{x}_n,\tilde{\boldsymbol{\Theta}}(\tau))\right]\left[\sum_{n=1}^{N}E(\mathbf{z}_n\mathbf{z}_n{}^H\mid\mathbf{x}_n,\tilde{\boldsymbol{\Theta}}(\tau))\right]^{-1}\\ \tilde{\boldsymbol{\Psi}}(\tau+1)&=\frac{1}{N}diag\left\{\sum_{n=1}^{N}\left[\mathbf{x}_n\mathbf{x}_n{}^H-\tilde{\mathbf{A}}(\tau+1)E[\mathbf{z}_n\mid\mathbf{x}_n,\tilde{\boldsymbol{\Theta}}(\tau)]\mathbf{x}_n^H\right]\right\}\end{aligned}\tag{9}$$

Where $\tilde{\boldsymbol{\Theta}}(\tau)$ is the parameters estimator of the τth iteration, *i.e.*, $\tilde{\boldsymbol{\Theta}}(\tau)=\{\tilde{\mathbf{A}}(\tau),\tilde{\boldsymbol{\Psi}}(\tau)\}$. After the algorithm converges, we can obtain the covariance $\boldsymbol{\Sigma}$ of the HRR complex samples $\mathbf{X}=[\mathbf{x}_n]_{n=1}^N$:

$$\boldsymbol{\Sigma}=\tilde{\mathbf{A}}\left(\frac{1}{N}\sum_{n=1}^{N}E[\mathbf{z}_n\mathbf{z}_n{}^H]\right)\tilde{\mathbf{A}}^H+\tilde{\boldsymbol{\Psi}}\tag{10}$$

Where $\tilde{\mathbf{A}}$, $\tilde{\boldsymbol{\Psi}}$ and $E[\mathbf{z}_n\mathbf{z}_n{}^H]$ are estimated by the EM algorithm.

B. Noise-robust Modification Algorithm

Several literatures [1-5,10-11] show that statistical recognition is an efficient method for the radar target recognition. There are two stages, *i.e.*, training stage and test stage, in a typical statistical recognition scheme. To deal with the target-aspect sensitivity, we usually divide data into frames, where the data in each frame are collected from a target-aspect sector roughly without the scatterers' motion through range cells (MTRC). Thus each frame is defined to be an aspect-frame from the target [5].We develop complex Gaussian models for HRR data from each aspect-frame. Let $\mathbf{x}^*$ be a test sample, since the frame models from a given target are assumed to be independent, the class-conditional likelihood of target c ($c\in\{1,2,\cdots,\mathrm{C}\}$) is $p(\mathbf{x}^*\mid c)=\max_m p(\mathbf{x}^*\mid m,c)$, with $m\in c\{1,2,\cdots,M_c\}$ denoting the frame index. We have $p(c\mid\mathbf{x}^*)\propto p(\mathbf{x}^*\mid c)p(c)$, according to Bayes theorem. where $p(c)$ is the prior class probability. If the targets have equal prior probabilities of occurrence, estimation of the posterior probability is turned into estimation of the class-conditional likelihood. Based on the complex Gaussian models, we can obtain $p(\mathbf{x}^*\mid m,c)=CG(\mathbf{x}^*\mid\mathbf{0},\boldsymbol{\Sigma})$, where $\boldsymbol{\Sigma}$ is estimated by the training samples with the above-mentioned models. Therefore, the class membership of $\mathbf{x}^*$ can be assigned to the class with the maximum posterior probability, $c^*=\arg\max_c p(c\mid\mathbf{x}^*)$.

In the above discussion, we assume the noise level of test samples is the same as that of training samples. However, the training samples can usually be obtained under high SNR via some cooperative measurement experiments; the test samples are usually got in non-cooperative circumstance, where the SNR cannot be guaranteed due to the large distance between non-cooperative targets and radar [1-5]. If we assume the measured test sample under low SNR can be expressed as $\hat{\mathbf{x}}^*=\mathbf{x}^*+\mathbf{n}$, where the noise level of $\mathbf{x}^*$ is the same as those of the training samples (namely with high SNR) and $\mathbf{n}$ is the new noise variable introduced from the test condition with the assumption that $\mathbf{n}\sim CG(\mathbf{n}\mid\mathbf{0},\sigma_n^2\mathbf{I}_D)$, then we have

$$p(\hat{\mathbf{x}}^*\mid m,c)=CG(\hat{\mathbf{x}}^*\mid\mathbf{0},\hat{\boldsymbol{\Sigma}})\tag{11}$$

With

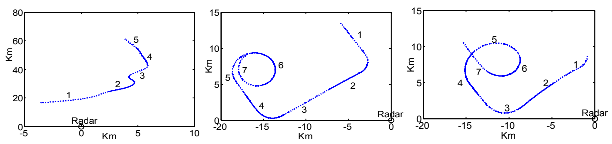
$$\begin{aligned}\hat{\boldsymbol{\Sigma}}&=E\left(\hat{\mathbf{x}}^*\hat{\mathbf{x}}^{*H}\right)=E\left((\mathbf{x}^*+\mathbf{n})(\mathbf{x}^*+\mathbf{n})^H\right)\\&=E\left(\mathbf{x}^*(\mathbf{x}^*)^H\right)+\sigma_n^2\mathbf{I}_P=\boldsymbol{\Sigma}+\sigma_n^2\mathbf{I}_P\end{aligned}\tag{12}$$

As shown in (12), our noise-robust modification algorithm is to modify the covariance with noise variance in the test stage. Since we can easily estimate the test noise level σ_n^2 via some preprocessing methods [5], the matched target indicator c^* and frame m^* can be easily obtained via the above frame matching method based on $\{p(\hat{\mathbf{x}}^*\mid m,c)\}_{m=1,c=1}^{M_c,C}$.

III. Experimental Results and Discussion

A. Measured Data

We examine the performance of the proposed models on the measured HRR data from three real airplanes (Yak-42, Cessna Citation S/II and An-26), of which the center frequency is 5520MHz and the bandwidth is 400MHz. The projections of target trajectories onto the ground plane are shown in Fig.1, from which the aspect angle of an airplane can be estimated according to its relative position to radar. As shown in Fig.1, the measured data are segmented, where each segment contains 26000 HRR samples and each sample is a 256-length vector, *i.e.*, $L=256$. Here data from the 2nd segment and a part of the 5th segment of Yak-42, the 6th and 7th segments of Cessna Citation S/II, the 5th and 6th segments of An-26 are taken as the training samples. We extract 5200 samples from the other data as test samples in our experiments. The length of each aspect-frame is $N=1024$. There are 35 aspect-frames for Yak-42, 50 for Cessna Citation S/II, and 50 for An-26. The original measured HRR data are collected under high SNR condition ($\text{SNR}\approx 40\text{dB}$).



(a) Yak-42 (b) Cessna Citation S/II (c) An-26

Fig. 1. Projections of target trajectories onto the ground plane.

B. Recognition Performance of Complex Gaussian Models

Here we compare the recognition performance of CAGC, CFA, AGC and FA. Table I shows the average correct recognition rates (ACRRs) of the four models on the original measured data.

TABLE I. ACRRS FOR DIFFERENT MODELS UNDER HIGH SNR CONDITION (%)

Models	**AGC**	**CAGC**	**FA**	**CFA**
ACRR	87.08	87.78	92.48	92.56

As shown in Table I, the performance of a complex Gaussian model is better than that of the corresponding real Gaussian model. This demonstrates that the phase information can be utilized to improve recognition performance. Besides, the recognition rates of FA and CFA are respectively higher than those of AGC and CAGC. We know AGC and CAGC models assume that HRR samples are statistically independent. However, there may be multiple scattering between scatterers from different range cells. Therefore, FA and CFA models, which consider the correlation between range cell echoes, match HRR data better than AGC and CAGC. Fig.2 shows that the HRR samples are not independent.

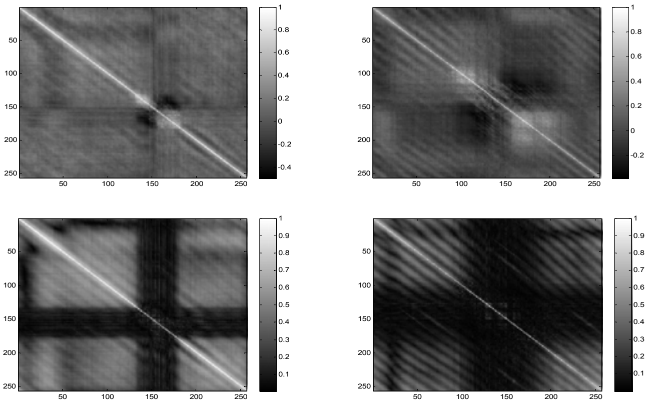


Fig. 2. Normalized covariance matrixes of measured data. The two in the first row come from HRR real-valued signals and the two come in the second second row come from HRR complex-valued signals.

C. Noise-robust Modification Algorithm Performance

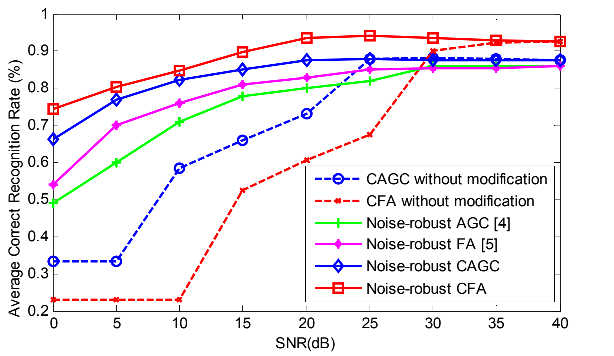


Fig. 3. Variation of the recognition performance with the test SNR. All these methods in this figure are trained with the training data under high SNR, but the recognition are implemented under different SNR conditions with random Gaussian white noise added to the test samples.

Fig.3 shows the variation of ACRRs with the test SNR. Since the noise-robust modification algorithm considers the noise component of the test data, it largely improves the recognition performance under the low test SNR. The recognition accuracies of the noise-robust CAGC and noise-robust CFA are significantly better than those of CAGC and CFA without modification under the low-SNR cases ($\mathrm{SNR} \le 25\mathrm{dB}$). Note that, when the SNR of test HRR complex samples is $5\mathrm{dB}$, CAGC and CFA without modification fail to recognize the samples, while the recognition rates of noise-robust CAGC and CFA are 77.09% and 80.25%, respectively. The recognition results of Gaussian models without modification are comparable to those of the modified Gaussian models under high-SNR cases ($\mathrm{SNR} \ge 30\mathrm{dB}$). Fig.3 also shows that our noise-robust CFA is always superior to the noise-robust FA [5], and the ACRRs of our noise-robust CAGC are also higher than those of the noise-robust AGC [4]. As the results show, the noise-robust modification algorithm for complex HRRP can yield better recognition performance than that for real HRRP, when the SNR is low.

IV. Conclusion

This paper demonstrates that initial phase does not affect the parameter estimators of complex Gaussian models, and takes CAGC and CFA for example to build models for HRR complex signals. In addition, we present a noise-robust modification algorithm in the complex domain, to improve the recognition performance under the low SNR condition. The experimental results show that the phase information can be utilized to improve the recognition performance, and the noise-robust modification algorithm can yield good recognition performance when the SNR is low.

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